Chapter 2 : Variable and Datatype.

A variable is the name given to a memory location in a program for example

a = 30
b = "Harry"
c = 71.22

→ Variables = Container to store a value.
→ Keywords = Reserved words in Python.
→ Identifiers = Class / function / variable name

Data Types:
Primarily there are following data types in python.

1. Integers.
2. Floating point numbers.
3. Strings.
4. Booleans.
5. None.

Python is a language that automatically identifies the type of data for us.

a = 71 => Identifies 'a' as class <int>
b = 88.44 => Identifies 'b' as class <float>
name = "Harry" => Identifies 'name' as class <string>

Rules for defining a variable name → Also applies to other identifiers

→ A variable name can contain alphabits, digits and underscore.

→ A variable name can ~~contain~~ only start with an alphebit and underscore.
→ A variable name can't start with a digit.
→ No white space is allowed to be used inside a variable name.
Examples: harry, one8, Seven_, _Seven etc.

Operators in Python :

1. Arthemetic ⇒ +, -, *, / etc
2. Assignment ⇒ =, +=, -=, etc.
3. Comparison ⇒ ==, >, >=, <, != etc.
4. logical ⇒ and, or not.

type () function and Typecasting

type, function is used to find the data type of a given variable in Python.

```
a = 31
type (a) => Class <int>

b = "31"
type (b) => class <string>
```

A number can be converted into a String and vice verse (if possible)

There are many function to convert one data type into another.

Str (31) => "31" => Integer to string Conver
int ("32") => 32 => String to integer Conversion
float (32) => 32.0 => Integer to float Conversion.

Here "32" is a string literal and 31 a number literal.

Input function.
This function allows the user to take task from the Keyword as a String.

a = input ("Enter name") => If a is "harry" the user entered harry.

It is important to note that => If a is "34" user entered 34.
the output of input is always a string (even if the number is entered)